Filter Functions for Dynamical Decoupling Sequences

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Abstract

We derive explicit filter functions $F(\omega, t)$ for the Ramsey, Hahn echo, CPMG-n, XY-8, and UDD-n pulse sequences. Each derivation follows from direct evaluation of the squared Fourier transform of the modulation function $f(t')$, as established in document 02. We characterize each filter function's spectral sensitivity and discuss its implications for noise spectroscopy.

1 Ramsey Sequence

For the Ramsey sequence there are no π pulses, so $f(t') = 1$ for all $t' \in [0, t]$. The filter function integral evaluates to:

$$\int_0^t e^{i\omega t'}\, dt' = \frac{1}{i\omega}\left(e^{i\omega t} - 1\right) \tag{1}$$

Taking the magnitude squared:

$$F_{\text{Ramsey}}(\omega, t) = \frac{1}{\omega^2}\left(e^{i\omega t} - 1\right)\left(e^{-i\omega t} - 1\right) = \frac{2 - 2\cos(\omega t)}{\omega^2} \tag{2}$$

Using the identity $1 - \cos\theta = 2\sin^2(\theta/2)$:

$$\boxed{F_{\text{Ramsey}}(\omega, t) = \frac{4\sin^2(\omega t/2)}{\omega^2}} \tag{3}$$

In the limit $\omega \to 0$, using $\sin(\omega t/2) \approx \omega t/2$:

$$F_{\text{Ramsey}}(\omega \to 0, t) = t^2 \tag{4}$$

The Ramsey filter function peaks at $\omega = 0$ with maximum value t^2. It is a **low-pass filter** — maximally sensitive to slow, DC-like noise and quasistatic fluctuations. This is why Ramsey dephasing in superconducting qubits is dominated by $1/f$ noise [3].

2 Hahn Echo Sequence

For the Hahn echo, a single π pulse is applied at $t/2$, so:

$$f(t') = \begin{cases} +1 & 0 \le t' < t/2 \\ -1 & t/2 \le t' \le t \end{cases} \tag{5}$$

The filter function integral splits into two parts:

$$\int_0^t f(t')\, e^{i\omega t'}\, dt' = \int_0^{t/2} e^{i\omega t'}\, dt' - \int_{t/2}^t e^{i\omega t'}\, dt' \tag{6}$$

Evaluating each integral:

$$\int_0^{t/2} e^{i\omega t'}\, dt' = \frac{1}{i\omega}\left(e^{i\omega t/2} - 1\right) \tag{7}$$

$$\int_{t/2}^t e^{i\omega t'}\, dt' = \frac{1}{i\omega}\left(e^{i\omega t} - e^{i\omega t/2}\right) \tag{8}$$

Combining:

$$\frac{1}{i\omega}\left(2e^{i\omega t/2} - 1 - e^{i\omega t}\right) \tag{9}$$

Factoring out $e^{i\omega t/2}$ and using $2 - e^{-i\omega t/2} - e^{i\omega t/2} = 2 - 2\cos(\omega t/2) = 4\sin^2(\omega t/4)$:

$$= \frac{e^{i\omega t/2}}{i\omega} \cdot 4\sin^2\left(\frac{\omega t}{4}\right) \tag{10}$$

Taking the magnitude squared (note $|e^{i\omega t/2}|^2 = 1$):

$$\boxed{F_{\text{Hahn}}(\omega, t) = \frac{16\sin^4(\omega t/4)}{\omega^2}} \tag{11}$$

In the limit $\omega \to 0$:

$$F_{\text{Hahn}}(\omega \to 0, t) \propto \omega^2 t^4 \to 0 \tag{12}$$

The Hahn echo filter function **vanishes at** $\omega = 0$, confirming that the π pulse refocuses slow noise completely. The filter peaks at:

$$\frac{\omega t}{4} = \frac{\pi}{2} \implies \omega_{\text{peak}} = \frac{2\pi}{t} \tag{13}$$

The Hahn echo is a **bandpass filter** centered at $\omega = 2\pi/t$. By varying t, the filter peak sweeps across the noise spectrum [1].

3 CPMG-n Sequence

CPMG-n consists of n equally spaced π pulses at times:

$$t_j = \frac{(2j-1)t}{2n}, \quad j = 1, 2, \ldots, n \tag{14}$$

The modulation function alternates sign at each pulse:

$$f(t') = (-1)^k, \quad t_k < t' < t_{k+1} \tag{15}$$

where $t_0 = 0$ and $t_{n+1} = t$. The filter function integral becomes a telescoping sum:

$$\int_0^t f(t')\, e^{i\omega t'}\, dt' = \sum_{k=0}^{n} (-1)^k \int_{t_k}^{t_{k+1}} e^{i\omega t'}\, dt' = \sum_{k=0}^{n} (-1)^k \frac{e^{i\omega t_{k+1}} - e^{i\omega t_k}}{i\omega} \tag{16}$$

Adjacent terms in the sum share boundary exponentials with opposite signs, leading to significant cancellation. After collecting terms, substituting $t_k = (2k-1)t/2n$, and summing the resulting geometric series, the sum evaluates to a closed form. Taking the magnitude squared gives [1]:

$$\boxed{F_{\text{CPMG-n}}(\omega, t) = \frac{16 \sin^4\left(\dfrac{\omega t}{4n}\right)}{\omega^2} \cdot \frac{\sin^2\left(\dfrac{\omega t}{2}\right)}{\sin^2\left(\dfrac{\omega t}{2n}\right)}} \tag{17}$$

For $n = 1$ this reduces exactly to $F_{\text{Hahn}}(\omega, t) = 16 \sin^4(\omega t/4)/\omega^2$, confirming consistency with the Hahn echo result.

The interference factor $\sin^2(\omega t/2)/\sin^2(\omega t/2n)$ is maximized when $\omega t/2n = \pi/2$, giving the approximate peak location:

$$\omega_{\text{peak}} \approx \frac{n\pi}{t} \tag{18}$$

By varying both n and t, the filter peak can be swept across the noise spectrum, enabling reconstruction of $S(\omega)$ [2]. Note that the true numerical peak is shifted slightly below $n\pi/t$ due to the $1/\omega^2$ denominator, consistent with the analogous shift observed for the Hahn echo.

4 XY-8 Sequence

XY-8 is a phase-cycled dynamical decoupling sequence consisting of 8 π pulses applied along alternating X and Y axes in the pattern X–Y–X–Y–Y–X–Y–X, with equally spaced pulse times identical to CPMG-8:

$$t_j = \frac{(2j-1)t}{16}, \quad j = 1, 2, \ldots, 8 \tag{19}$$

For dephasing noise (Z-axis noise), the modulation function $f(t')$ is identical to CPMG-8, since both X and Y rotations flip the sign of σ_z phase accumulation equally. Therefore:

$$\boxed{F_{\text{XY-8}}(\omega, t) = F_{\text{CPMG-8}}(\omega, t) = \frac{16 \sin^4\left(\frac{\omega t}{32}\right)}{\omega^2} \cdot \frac{\sin^2\left(\frac{\omega t}{2}\right)}{\sin^2\left(\frac{\omega t}{16}\right)}} \tag{20}$$

The filter peaks at $\omega_{\text{peak}} \approx 8\pi/t$.

The advantage of XY-8 over CPMG-8 lies not in the filter function but in **robustness to pulse errors**. Imperfect π pulses leave residual rotations that accumulate coherently in CPMG. The alternating X/Y phase cycling in XY-8 causes these residual errors to cancel to first order, significantly improving performance on real hardware [4].

Note on pulse error suppression

The filter function formalism assumes ideal π pulses. On IBM hardware, gate fidelities are finite and systematic pulse errors are present. For this reason, XY-8 is preferred over CPMG-8 for hardware experiments in this project, even though their filter functions are theoretically identical.

5 UDD-n Sequence

Uhrig Dynamical Decoupling (UDD) [5] places n π pulses at nonuniform times determined by:

$$t_j = t \cdot \sin^2\left(\frac{j\pi}{2(n+1)}\right), \quad j = 1, 2, \ldots, n \tag{21}$$

This specific spacing is derived by requiring that the coherence decay $\chi(t)$ vanishes to order n in t for a generic smooth $S(\omega)$. Uhrig showed that this is the optimal pulse placement for suppressing dephasing with exactly n pulses [5], in the sense that no other n-pulse sequence can achieve higher-order cancellation.

Filter Function

The filter function follows the same general sum as CPMG:

$$\tilde{F}(\omega, t) = \sum_{k=0}^{n} (-1)^k \frac{e^{i\omega t_{k+1}} - e^{i\omega t_k}}{i\omega} \tag{22}$$

with $t_0 = 0$, $t_{n+1} = t$, and $F_{\text{UDD}}(\omega, t) = |\tilde{F}(\omega, t)|^2$.

Unlike CPMG, the nonuniform pulse spacing prevents summation as a geometric series. **No closed-form expression exists for $F_{\text{UDD}}(\omega, t)$** — the filter function must be evaluated numerically for each (ω, t, n).

Comparison with CPMG

UDD is optimal for **smooth** $S(\omega)$ — noise spectra without sharp features. In this regime, UDD achieves the same suppression order as CPMG with fewer pulses, making it more efficient.

However, for noise with sharp spectral features — such as Lorentzian peaks from TLS defects — CPMG can outperform UDD because its uniform spacing produces a filter function with better-defined harmonic structure that can be tuned to avoid TLS resonances [1].

In this project, both sequences are implemented and their reconstructed $S(\omega)$ are compared, with the expectation that they agree for smooth noise and diverge near TLS peaks.

6 Summary

Sequence	**Filter function**	**Peak frequency**	**Character**
Ramsey	$4\sin^2(\omega t/2)/\omega^2$	$\omega = 0$	Low-pass
Hahn echo	$16\sin^4(\omega t/4)/\omega^2$	$\omega = 2\pi/t$	Bandpass
CPMG-n	$16\sin^4(\omega t/4n)\sin^2(\omega t/2)/[\omega^2\sin^2(\omega t/2n)]$	$\omega \approx n\pi/t$	Tunable bandpass
XY-8	Same as CPMG-8	$\omega \approx 8\pi/t$	Phase-robust bandpass
UDD-n	No closed form	Nonuniform	Optimal for smooth $S(\omega)$

Table 1: Filter functions for dynamical decoupling sequences used in this project.

References

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